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INTRODUCTION TO INTRUSION DETECTION SYSTEM

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INTRODUCTION TO INTRUSION DETECTION SYSTEM

OUTLINE

- Terminology
- Intrusion Detection System
- Advantages of IDS
- IDS Working Concept

TERMINOLOGY

WHAT IS AN INTRUSION?

ANY UNAUTHORIZED OR MALICIOUS ACTIVITY THAT TRIES TO ACCESS, MANIPULATE, DISRUPT, OR DAMAGE A COMPUTER SYSTEM, NETWORK, OR DATA.

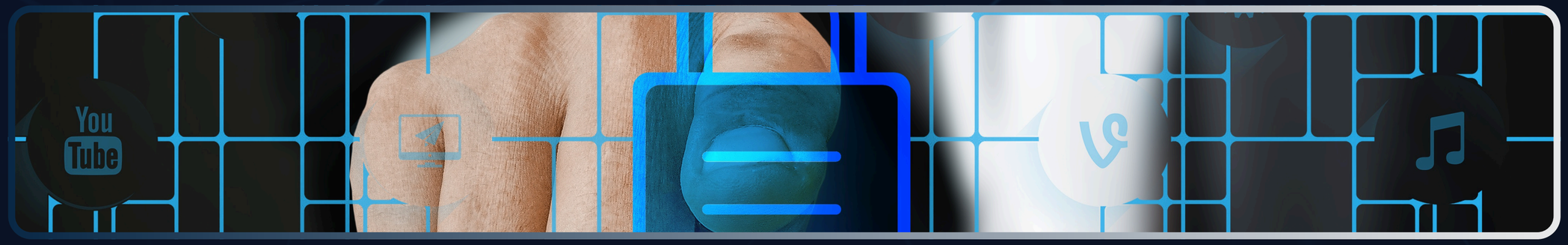
SIGN OF INTRUSION

- Unaccountable disk utilization
- Unaccountable file system modification
- Unaccountable CPU utilization
- Network Saturation
- Unknown process using sockets
- Abnormal network /system activity

INTRUSION DETECTION SYSTEM

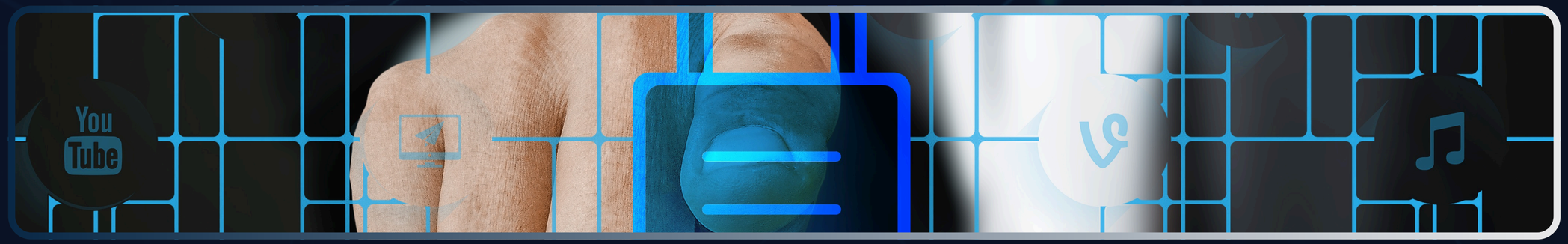
DEFINITION

Its a Security Monitoring Tool. It **Gathers** and **analyzes** information from within a computer or a network, to identify possible violations of security policy, including unauthorized access, as well as misuse.



PURPOSE

The main goal of an IDS is detection. It alerts administrators if something suspicious happens. IDS primarily monitors and reports.



HOW IT WORKS?

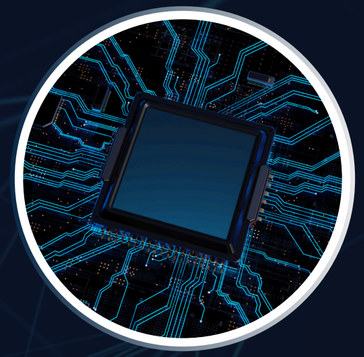
- Think of IDS like a security camera for your IT systems. It doesn't physically stop an intruder, but it records, recognizes, and warns you when someone is breaking in.

- An Intrusion Detection System (IDS) works by monitoring and analyzing network traffic or system activities to identify signs of malicious behavior or policy violations. It uses signature-based detection and anomaly-based detection. When something suspicious is detected, the IDS generates alerts to notify administrators so they can respond.

THREATS

INTRUSION

MALICIOUS CODE



DENIAL-OF-SERVICE

MAILBOMBING

SPAMMING

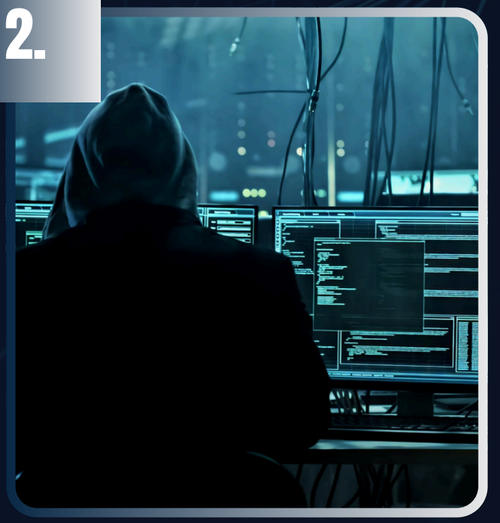
FORGERY

TYPE OF INTRUDERS

1.

INSIDERS

2.



OUTSIDERS

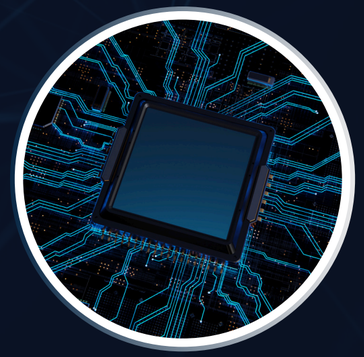
INSIDERS

- Intruders that legitimately use your internal network
- Include users who misuse privileges or who impersonate higher privileged users
- A frequently quoted statistic is that 80% of security breaches are committed by insiders

COMMON INSIDER ATTACK

DATA THEFT BY EMPLOYEE

PRIVILEGE MISUSE



SABOTAGE

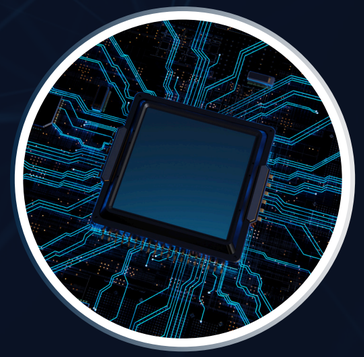
OUTSIDERS

- From outside network
- May attack your external presence (deface web servers, forward spam through e-mail servers)
- May come from Internet, dial-up lines, physical break-ins, or from partner (vendor, customer, reseller, etc) network that is linked to your corporate network

COMMON OUTSIDER ATTACK

PHISHING ATTACK

RANSOMWARE ATTACK



SQL INJECTION

WAYS INTRUDER GET INTO SYSTEM

Physical Intrusion

- Intruders have physical access to a machine

System Intrusion

- Assumes the intruder already has a low-privilege user account on the system

Remote Intrusion

- Involves a intruder who attempts to penetrate a system remotely across the network

GROUP ASSIGNMENT

CONT...

WHY DO I NEED IDS, I HAVE FIREWALL?

NOT ALL TRAFFIC MAY GO THROUGH A FIREWALL

Users, for various reasons ranging from ignorance to impatience, sometimes set up unauthorized modem connections between their systems that are connected to the network and outside Internet service providers. The firewall cannot mitigate risk associated with connections it neversees.

NOT ALL THREATS ORIGINATES FROM OUTSIDE.

- Firewalls limit access between networks to prevent intrusion and do not signal an attack from inside the network. An IDS evaluates a suspected intrusion once it has taken place and signals an alarm. An IDS also watches for attacks that originate from within a system.

- The firewall only sets barriers between the internal network and the Internet. If the traffic reflecting security breaches never passes the firewall, it cannot detect the problem.

- Positioning an IDS inside the firewall helps you understand attacks that originate outside your network. It may not, however, cover exploits that originate from inside your network targeting your hosts, depending on your network's topology.

FIREWALL DOES NOT PROTECT APPROPRIATELY AGAINST APPLICATION LEVEL WEAKNESSES AND ATTACKS

- Application-based attacks refer to the practice of exploiting vulnerabilities in applications by sending packets that communicate directly with those applications.
- Therefore, one could exploit a problem with Web software by sending an HTTP command that exercises a buffer overflow in the webapplication.
- If the firewall is configured to pass HTTP traffic, the packet containing the attack will pass.

FIREWALLS ARE SUBJECT TO ATTACKS THEMSELVES, WHICH SUBJECT TO MANY ATTACKS

- A common strategy is to use tunneling to bypass firewall protections.
- Tunneling is the practice of encapsulating a message in one protocol that might be blocked by firewall filters, inside a second message.

REAL LIFE SCENARIO

ADVANTAGE & DISADVANTAGE OF IDS

ADVANTGAGE

Early Threat Detection

- Alerts administrators of potential attacks in real time or near real time.
- Enables quick response before serious damage occurs.

Monitors System and Network Activity

- Continuously inspects traffic, logs, and behavior patterns.
- Helps identify misuse or suspicious changes.

Detects Known and Unknown Attacks

- Signature-based IDSs detect known threats.
- Anomaly-based IDSs can flag unknown or novel threats.

DISADVANTAGE

High False Positives

- Especially in anomaly-based IDS; normal behavior might be flagged as malicious.
- Can lead to alert fatigue.

Reacts to attacks rather than preventing them

- IDS is deployed passively to monitor network traffic.
- It’s not inline with the network traffic, so it cannot block packets.

Cannot deal with encrypted network traffic

- Read malicious scripts or payloads hidden inside HTTPS.
- Detect command-and-control instructions sent inside SSL.
- Identify sensitive data exfiltration via encrypted channels

GROUP ASSIGNMENT 2

1. PACKET DECODER

The packet decoder takes packets from different types of network interfaces and prepares the packets to be preprocessedor to be sent to the detection engine.

2. PREPROCESSORS

- Components or plug-in that can be used with Snort to arrange or modify data packets before the detection engine does some operation to find out if the packet is being used by an intruder.
- Important for any IDS to prepare data packets to be analyzes against rules in the detection engine.

3. DETECTION ENGINE

1. To detect if any intrusion activity exists in a packets.
2. The load of the detection engine depends upon the following factors:

- Number of rules
- Power of the machine on which Snort is running
- Speed of internal bus used in the Snort machine
- Load of the network

4. LOGGING AND ALERTING SYSTEM

- Logs are kept in simple text files, tcpdump- style files or some other form.
- All of the log files are stored under **/var/log/snor**t folder by default

5. OUTPUT MODULES

- Control the type of output generated by the logging and alerting system

CONCLUSION

- Intrusion Detection Systems (IDS) play a crucial role in modern cybersecurity by detecting and alerting on potential threats and policy violations.
- IDS complements other security mechanisms like firewalls and Intrusion Prevention Systems (IPS) as part of a Defense-in-Depth strategy.
- While IDS helps identify attacks, it is not a complete security solution—proper configuration, regular updates, and additional layers of protection are essential.
- Understanding how IDS works, its strengths, and its limitations will help you design more secure networks and respond effectively to security incidents.

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THANK YOU